

Product Carbon Footprint Study Report

Product Name: hard disk drive production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: hard disk drive production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 item hard disk drive production, for desktop computer

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 item hard disk drive production, for desktop computer.

2. System Boundary

hard disk drive production, for desktop computer, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of hard disk drive production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **14.885 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

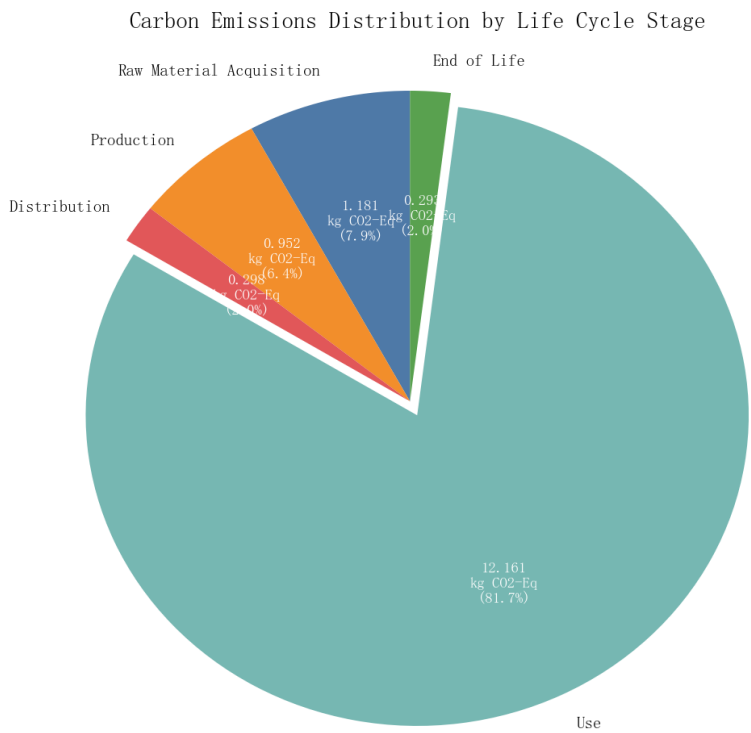


Table 1: Carbon Emissions by Life Cycle Stage

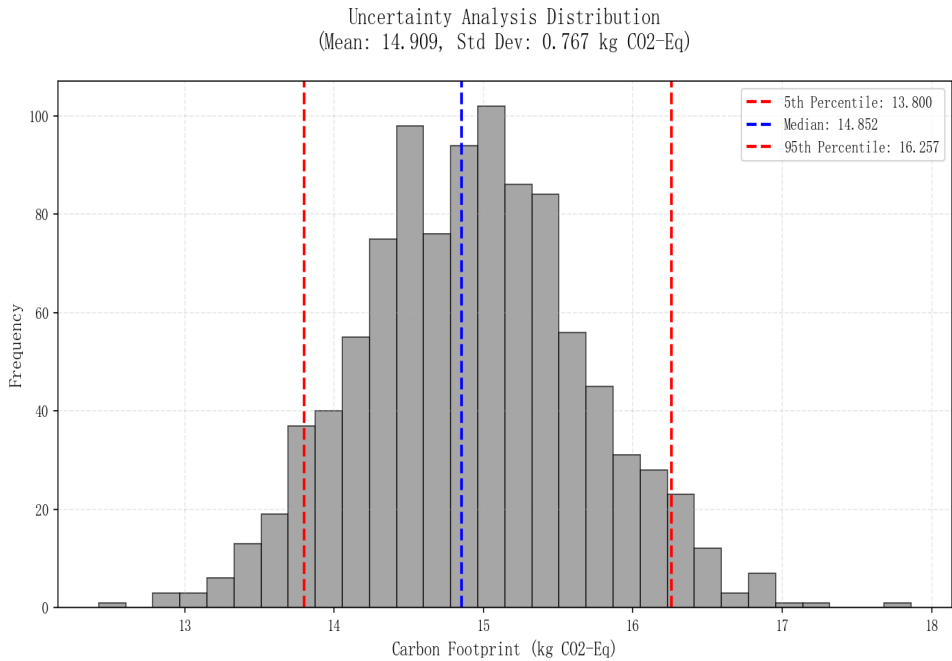
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	1.181	7.94
Production	0.952	6.39
Distribution	0.298	2.00

Use	12.161	81.70
End of Life	0.293	1.97
Total	14.885	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	14.909
Standard Deviation	0.767
Coefficient of Variation (CV)	5.14%
Median	14.852
5th Percentile	13.800
95th Percentile	16.257

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the hard disk drive (HDD) production is calculated at 14.885 kg CO₂-Eq per unit. This value is moderate for an electronic component, reflecting a product whose environmental impact is dominated not by manufacturing but by its operational energy consumption. The lifecycle stage analysis reveals a highly skewed contribution pattern: the use phase accounts for 81.7% (12.161 kg CO₂-Eq) of the total impact, while raw material extraction, production, distribution, and end-of-life together contribute only 18.3%. This distribution is typical for energy-using products where electricity consumption over the product's lifespan far outweighs the embodied emissions.

The predominant emission source is therefore the electricity consumed during the HDD's operational life. The raw material stage (7.9%) and production stage (6.4%) represent secondary contributions, likely driven by the energy and material intensity of manufacturing components such as platters, motors, and circuit boards. Distribution and end-of-life are negligible (2.0% each). The "Unknown" top processes (all zero) indicate that the study lacks granularity in identifying specific high-impact processes, which may mask additional hotspots. Overall, the results highlight that any effective carbon reduction strategy must prioritize the use phase, followed by improvements in material sourcing and manufacturing efficiency.

2. Statement of Assumptions and Limitations

This study relies on several key assumptions that shape the results. The use stage assumes a typical operational profile, including average daily usage hours, a lifespan of approximately 5 years, and an electricity grid mix representative of the target market (e.g., global average or a specific regional mix). The raw material and production stages assume industry-average emission factors for metals, plastics, and electronic components, without product-specific supplier data. The distribution stage assumes a standard transport mode (e.g., truck and ocean freight) and distances based on a generic global supply chain.

Limitations in data sources and quality are evident from the "Unknown" top processes, which indicate that process-level contributions for certain manufacturing steps (e.g., precision coating, motor assembly) were either not available or aggregated into higher-level categories. This leads to potential underestimation of manufacturing-related impacts. The system boundary is cradle-to-grave, covering raw material extraction, production, distribution, use, and end-of-life. However, it excludes capital goods (e.g., factory construction) and ancillary services (e.g., packaging). The uncertainty analysis (CV 5.14%, 90% CI 13.800–16.257 kg CO₂-Eq) suggests moderate confidence, but the confidence interval width (~2.5 kg CO₂-Eq) indicates that the true value could be up to 17% higher or lower than the mean.

3. Suggestions for Improvement

- ****Improve use-phase energy efficiency:**** Implement advanced power management features (e.g., idle spin-down, variable RPM) and promote energy-efficient drive designs to reduce the 81.7% use-phase contribution. Encourage end-users to adopt renewable energy sources for data centers and personal computers.

- ****Reduce raw material impacts:**** Substitute high-carbon materials (e.g., virgin aluminum for platters) with recycled or low-carbon alternatives. Collaborate with suppliers to source rare-earth magnets from certified sustainable mines with lower emission footprints.
- ****Enhance production process data quality:**** Replace “Unknown” top processes with primary data from HDD manufacturers. Conduct detailed process-level LCAs for key subassemblies (e.g., head stack assembly, spindle motor) to identify and mitigate hidden emission hotspots.
- ****Extend product lifespan:**** Design for durability and repairability to lengthen the use phase, thereby amortizing manufacturing emissions over a longer period. Offer firmware updates to maintain performance and delay replacement.
- ****Optimize distribution logistics:**** Consolidate shipments and shift to lower-carbon transport modes (e.g., rail instead of air freight) to further reduce the already low distribution impact (2.0%) and set a benchmark for supply chain decarbonization.